GENERIC POINTS OF IRREDUCIBLE COMPONENTS OF A DECENT SPACE EXIST AND ARE DECENT

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Generic points after etale cover. Morphisms of Spaces, lemma generic-point-exists (cross-chapter, named). Generic points exist after an etale surjection.
- Decency pullback. lemma-etale-named-properties (carded L26706). Decency of the pullback fibre points.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2878–2878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ABU: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1432–1432. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 03K6: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2915–2915. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NA: focused-seeds/geometry/S9-decent-decent-generic-points/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $x \in |X|$. The following are equivalent*

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *x is a generic point of an irreducible component of $|X|$,*
(2) *for any étale morphism $(Y, y) \to (X, x)$ of pointed algebraic spaces, y is a generic point of an irreducible component of $|Y|$,*
(3) *for some étale morphism $(Y, y) \to (X, x)$ of pointed algebraic spaces, y is a generic point of an irreducible component of $|Y|$,*
(4) *the dimension of the local ring of X at x is zero, and*
(5) *x is a point of codimension 0 on X*

Proof. Conditions (4) and (5) are equivalent for any algebraic space by definition, see Properties of Spaces, Definition 04NA. Observe that any Y as in (2) and (3) is decent by Lemma 0ABU. Thus it suffices to prove the equivalence of (1) and (4) as then the equivalence with (2) and (3) follows since the dimension of the local ring of Y at y is equal to the dimension of the local ring of X at x. Let $f : U \to X$ be an étale morphism from an affine scheme and let $u \in U$ be a point mapping to x.

Assume (1). Let $u' \rightsquigarrow u$ be a specialization in U. Then $f(u') = f(u) = x$. By Lemma 03K5 we see that $u' = u$. Hence u is a generic point of an irreducible component of U. Thus $\dim(\mathcal{O}_{U,u}) = 0$ and we see that (4) holds.

Assume (4). The point x is contained in an irreducible component $T \subset |X|$. Since $|X|$ is sober (Proposition 03K6) we T has a generic point x'. Of course $x' \rightsquigarrow x$. Then we can lift this specialization to $u' \rightsquigarrow u$ in U (Lemma 03IL). This contradicts the assumption that $\dim(\mathcal{O}_{U,u}) = 0$ unless $u' = u$, i.e., $x' = x$. □